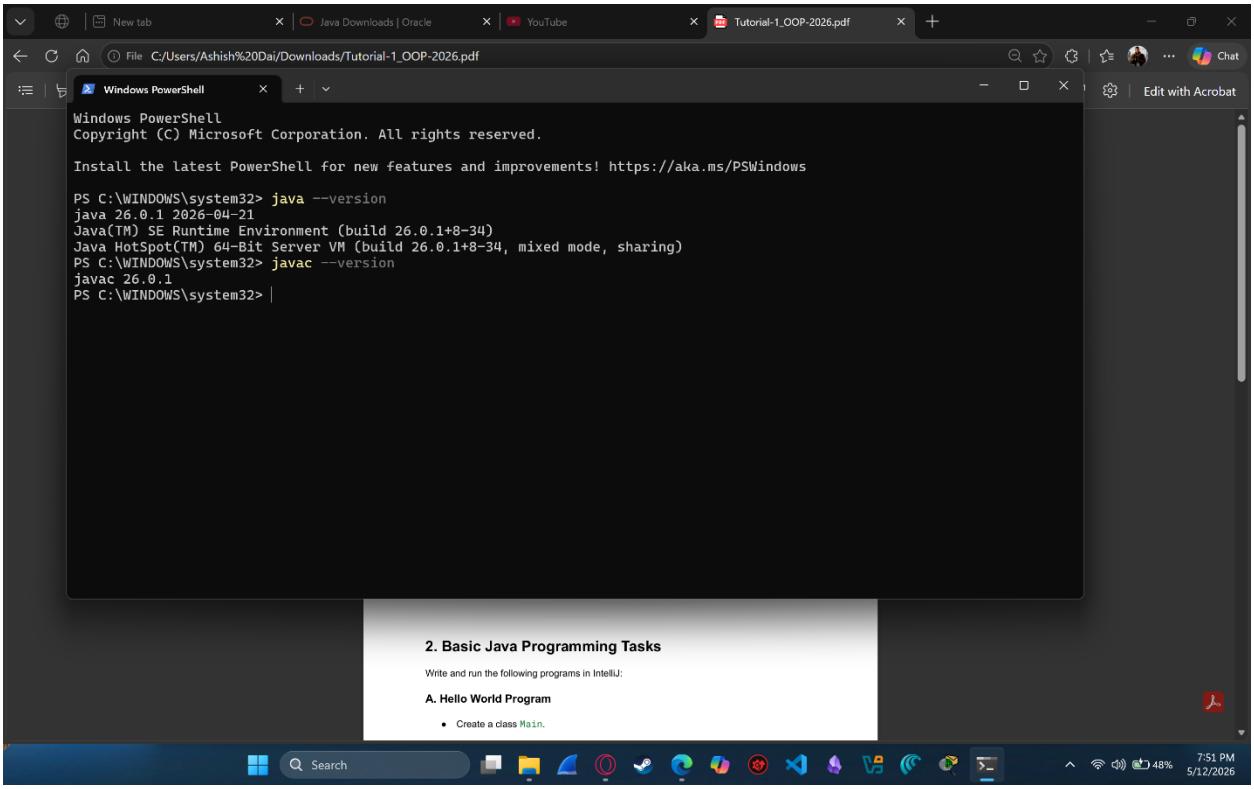
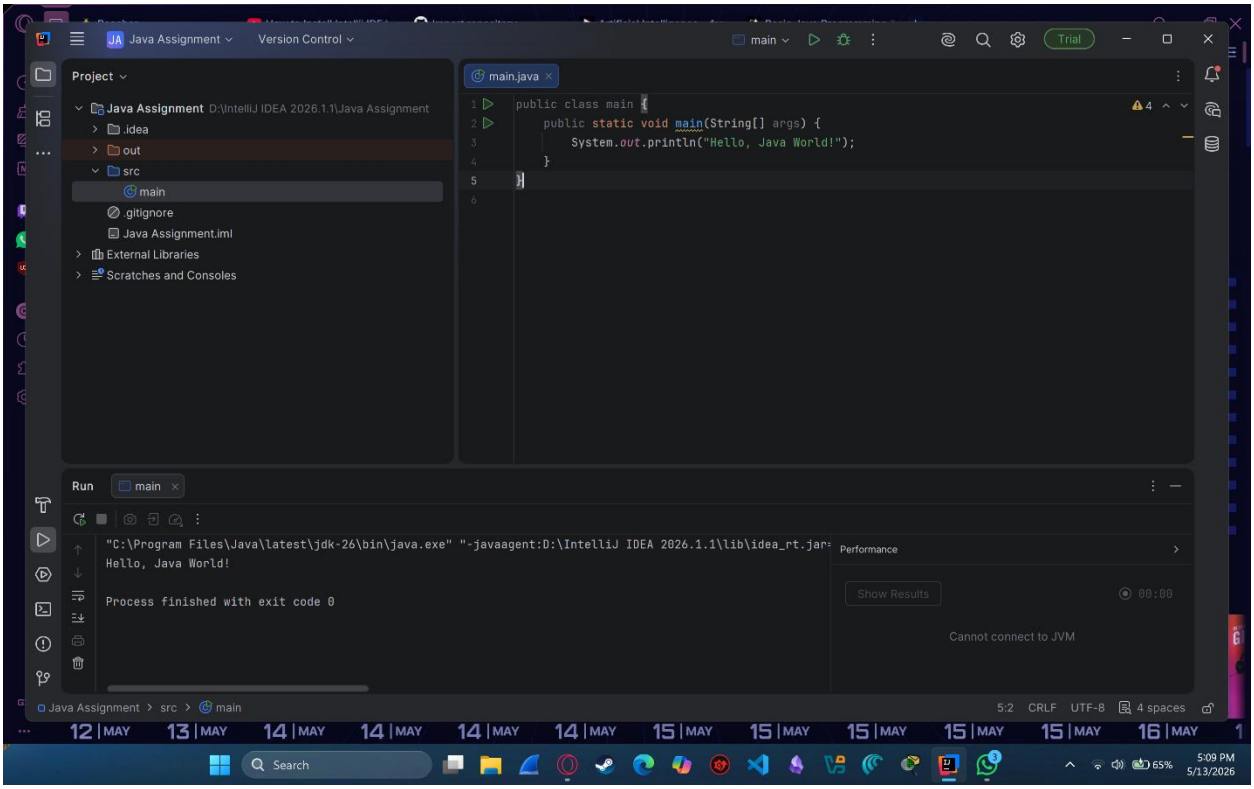


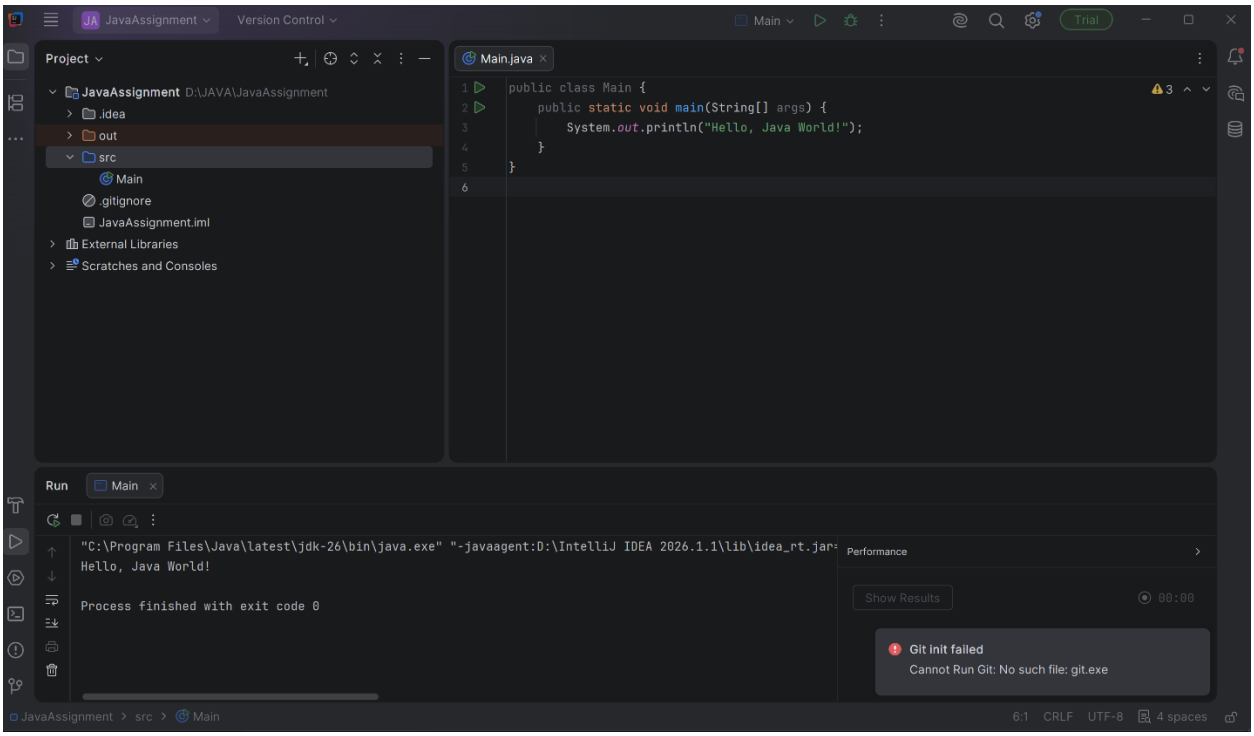
Screenshot 1: Java Installation Verification

This screenshot displays the successful installation of the Oracle JDK on my local machine. By running the java --version and javac --version commands in the terminal, I have verified that the system environment variables are correctly configured to run and compile Java applications.



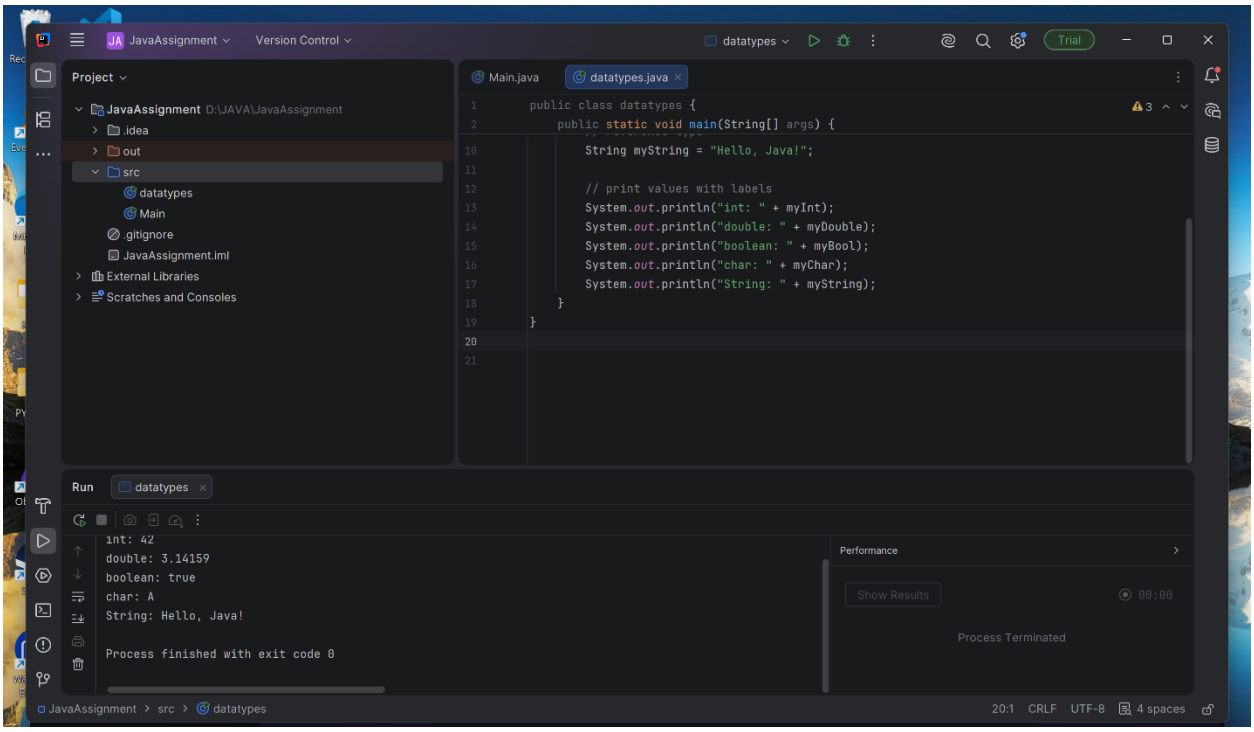
Screenshot 2: IDE Setup and Project Workspace

This screenshot demonstrates that IntelliJ IDEA Community Edition has been successfully installed and configured with the Java Development Kit (JDK). The workspace shows the newly created project for the tutorial tasks, displaying the project directory structure and the initialized classes ready for development.



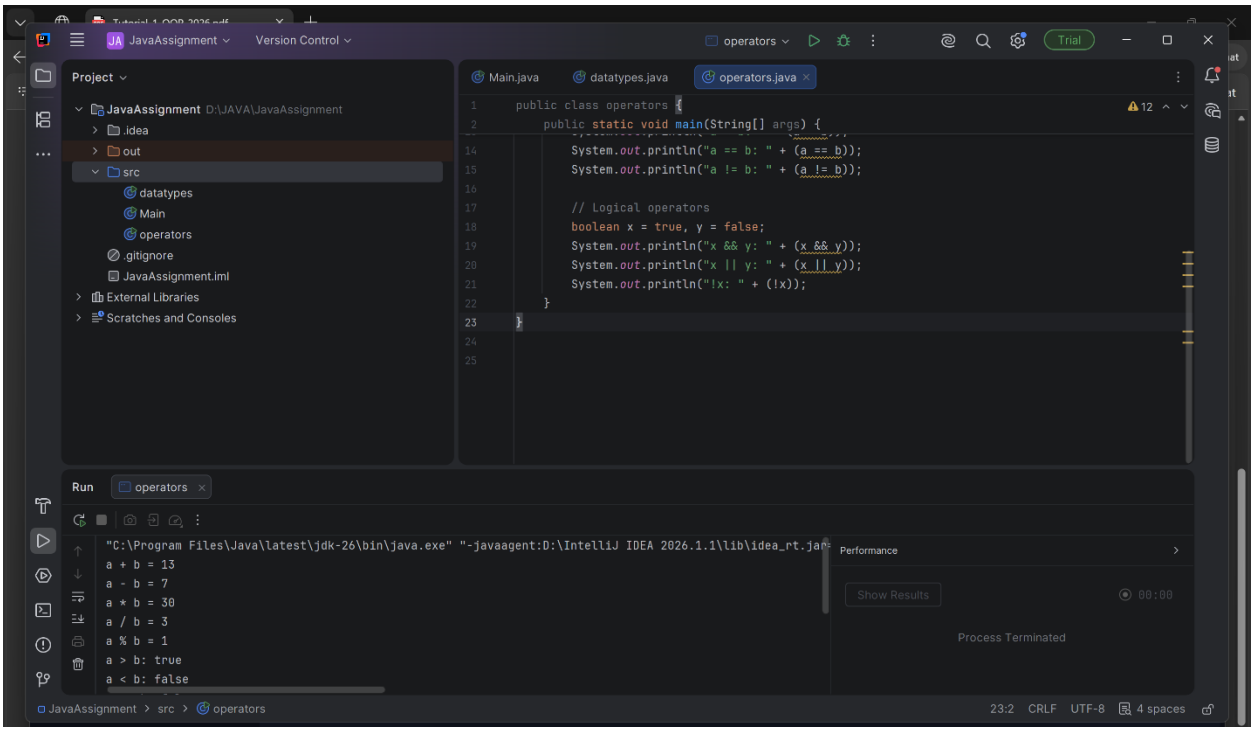
Screenshot 3: Hello World Execution

This screenshot displays the execution of the Main class. The console output confirms the successful printing of "Hello, Java World!" which verifies that the basic program structure and output stream are functioning correctly in the IntelliJ environment.



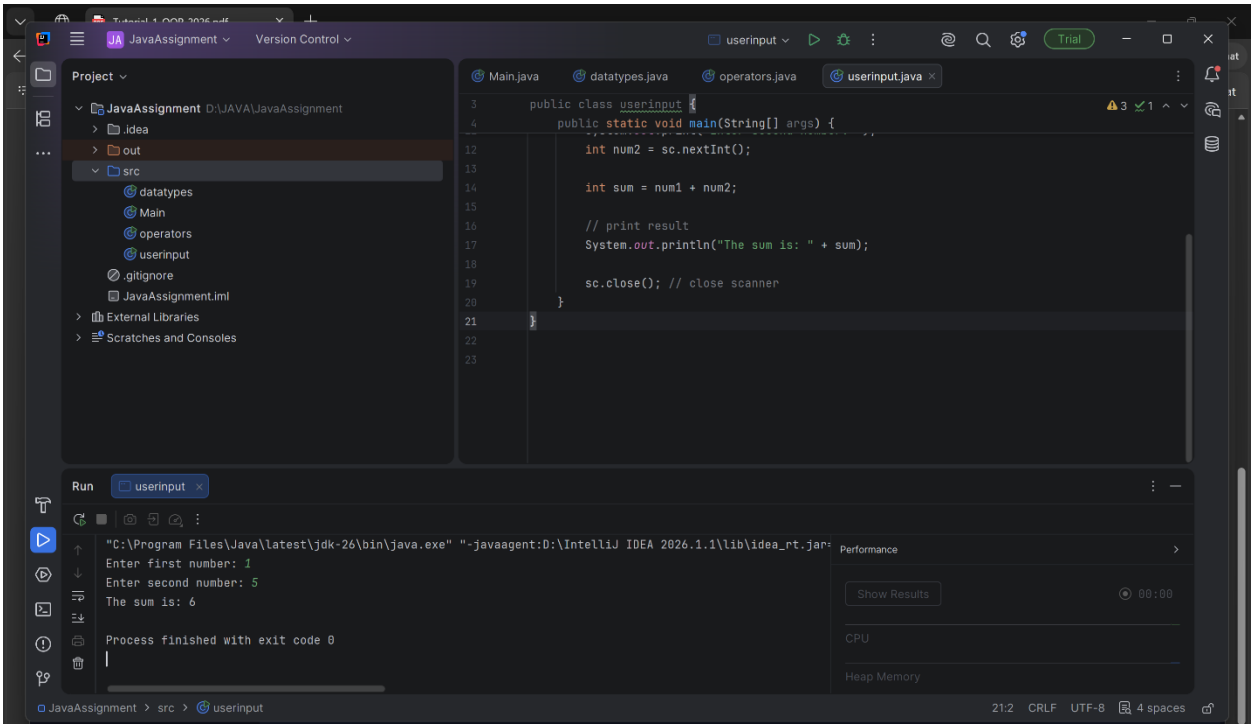
Screenshot 4: Variables and Data Types Output

This screenshot demonstrates the program demonstrating various Java data types, including int, double, boolean, char, and String. The console output shows the assigned values for each variable, confirming successful initialization and data handling.



Screenshot 5: Operators Program Results

This screenshot captures the results of the Operators program, showcasing the use of Arithmetic (+,—,*,/,%), Relational (>,<,==,!=), and Logical(&&,||,!) operators. The output proves that the mathematical and comparison logic is executing accurately within the Java compiler.



Screenshot 6: User Input

This screenshot shows the execution of the bonus task using the Scanner class to take two numerical inputs from the user. The console output displays the final sum of the two numbers, demonstrating proficiency in handling dynamic user input and basic arithmetic operations

1. Hello World Execution

```
public class Main{
  public static void main(String[] args){
      System.out.println("Hello, Java World!");
  }
}
```

Output:
Hello, Java World!

2. Variables and Data Types Output

```
public class datatypes{
    public static void main(String[] args){
      int myInt = 42;
      double myDouble = 3.14159;
      boolean myBool = true;
      char MyChar = 'A';
    String myString = "Hello, Java!"

System.out.println("int: " + myInt);
System.out.println("double: " + myDouble);
System.out.println("boolean: " + myBool);
```

```
System.out.println("char:" + myChar);
System.out.println("String:" + myString);
    }
}
```

Output:
int: 42
double: 3.14159
boolean: true
Char: A
String: Hello, Java!

3. Operators Program Results

```
public class operators {
    public static void main(String[] args) {
        // Arithmetic operators
        int a = 10, b = 3;
System.out.println("a+b =" + (a+b));
System.out.println("a-b =" + (a-b));
System.out.print ln("a*b =" + (a*b));
System.out.println("a/b =" + (a/b));
System.out.print ln("a%b =" + (a%b));
```

```
// Relational operators

System.out.println("a>b:"+(a>b));
System.out.println("a>b:"+ (a<b));
System.out.println("a==b:"+(a==b));
System.out.println("a!=b:"+(a!=b));

// Logical operators
 boolean x= true, y= false;

System.out.println("x && y:"+(x && y));
System.out.println("x || y:"+(x || y));
System.out.println("!x:"+(!x));
 }
}
```

Output:

a+b = 13	a==b*: false
a-b = 7	a!=b : true
a*b = 30	x && y: false
a/b = 3	x || y : true
a%b = 1	!x: false
a >b: True	
a< b: False	

4. User Input

```
import java.util.Scanner;

public class userinput {
    public static void main(String[] args){
    Scanner sc = new Scanner(System.in);
    //prompt user
    System.out.print("Enter first number:");
      int num1 = sc.nextInt();
System.out.print("Enter second number:");
  int num2= sc.nextInt();

int sum= num1 + num2;
//print result
System.out.println("The sum is:"+sum);
 sc.close();
    }
}
```

Output:

Enter first number: 3

Enter second number: 4

The sum is: 7